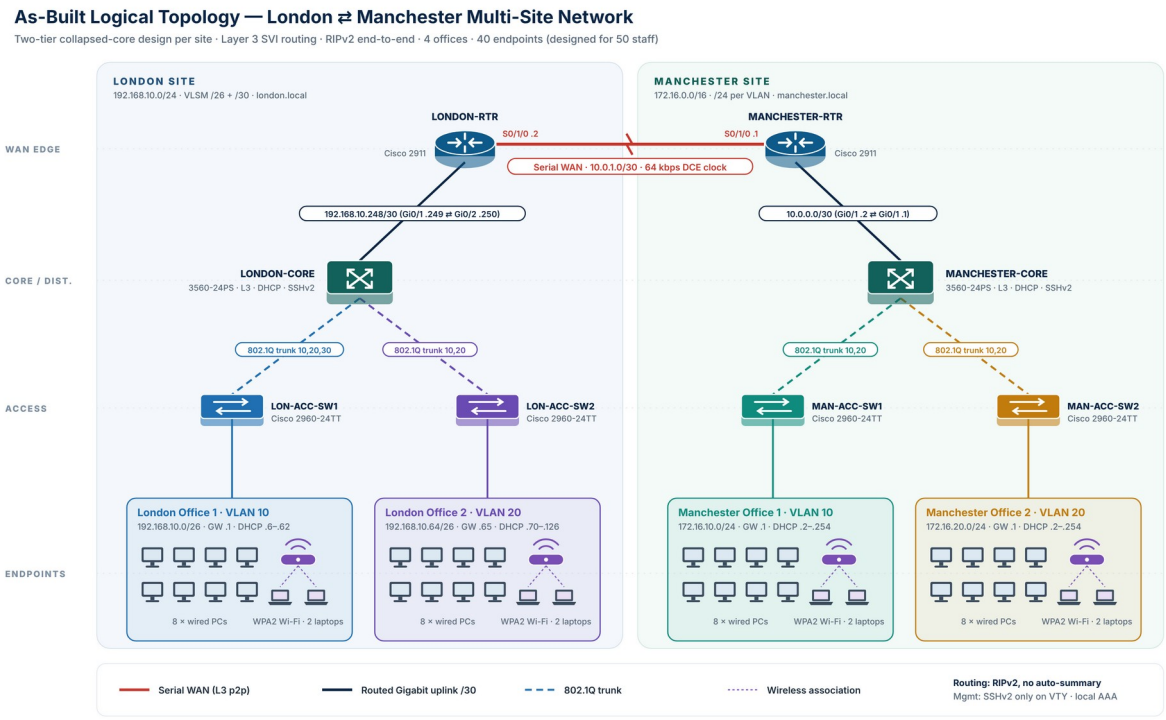
ENTERPRISE NETWORK DESIGN · CASE STUDY

Multi-Site Enterprise Network

London ⇄ Manchester: design, implementation and verification

Hierarchical LAN design, VLSM addressing, inter-VLAN routing on Layer 3 switches, RIPv2 dynamic routing over a serial WAN, centralised DHCP and SSH-only device management. Built and tested in Cisco Packet Tracer 8.2.



2 sites · 4 offices
London & Manchester, one routed domain

8 + 4 + 40
network devices · APs · endpoints

7 subnets
VLSM /26 · /24 · /30

RIPv2
classless, no auto-summary, 3-hop path

SSH only
Telnet off · local AAA on every VTY

7 / 7 tests
ping, path, SSH, routing, DHCP passed

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Version 1.0 · October 2026

Executive Summary

This case study documents the end-to-end design, build and verification of a secure, routed network for a fictional company with 50 staff across four offices: two in London and two in Manchester. Each city runs its own Layer 3 core that routes between office VLANs and hands out addresses by DHCP. The two cities are joined by a serial WAN link, and RIPv2 carries every subnet across it. Management access to all eight network devices is limited to SSH with local authentication.

The network was built in Cisco Packet Tracer 8.2 using real Cisco IOS command syntax on 2911 routers, 3560 multilayer switches and 2960 access switches. All seven verification tests passed. They covered same-site and inter-city reachability, packet-path tracing, SSH in both directions, routing-table convergence and DHCP leasing.

The report goes beyond the lab build. Section 7 audits the running configurations against hardening benchmarks and records nine concrete gaps. Section 9 turns the design into a target production architecture with a phased roadmap, so the lab design can be compared directly with how the same client would be built in production.

Table 1. *Results at a glance*

Area	Outcome
Architecture	Two-tier collapsed-core per site (router → L3 core → access), right-sized for 50 users
Addressing	RFC 1918 plan: VLSM /26 user LANs in London, /24 per VLAN in Manchester, /30 on all three point-to-point links
Segmentation	4 user VLANs (VLAN 10/20 per city), VLAN 30 reserved for growth; 802.1Q trunks pruned to required VLANs
Routing	RIPv2 with `no auto-summary` on 4 L3 nodes; every remote prefix learned (metrics 1–3)
Services	DHCP pools on both cores with gateway exclusions; WPA2-PSK Wi-Fi per office
Security	SSH-only VTY (v2 enforced on 6 of 8), local AAA, enable secret, banners; 9 gaps identified with fixes
Verification	7/7 tests passed (ICMP, simulation path, SSH ×2, `show ip route` ×2, DHCP)

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1. Introduction and Scope

Organisations that are spread across sites need a network that keeps each office fast and isolated locally while still letting the offices work as one. The client in this project is a fictional company with 50 staff in four offices, two per city. They need wired and wireless access, automatic addressing, secure remote administration and reliable communication between the two cities.

1.1 Aim

To design, implement and verify a secure, scalable and well-documented multi-site network that meets the client requirements below, and to show how the design would develop into a production-grade architecture.

1.2 Requirements and how they were met

Table 2. *Client requirements traceability*

ID	Requirement	Implementation	Evidence
R1	Hierarchical, scalable topology	Router → L3 core (3560) → access (2960) at each site	Fig. 1, §3
R2	Logical separation per office	VLAN 10 / VLAN 20 per city; VLAN 30 reserved	§4, Table 6
R3	Automatic IP addressing	DHCP pools on both cores, gateways excluded	§5.2, Figs. 14–15
R4	Dynamic inter-site routing	RIPv2, classless, across a serial WAN	§5.4, Figs. 12–13
R5	Secure device management	SSHv2, `login local`, `transport input ssh`	§5.5, Figs. 10–11
R6	Wired + wireless access	8 PCs + AP with 2 laptops per office (WPA2-PSK)	Figs. 2–4
R7	Proven cross-site connectivity	ICMP, simulation trace and SSH between cities	§6

1.3 Scope boundaries

- **In scope:** LAN/WAN design, addressing, VLANs, routing, DHCP, management security and verification.
- **Out of scope for the lab, covered as recommendations:** internet edge and NAT, firewalls, redundancy (dual WAN, HSRP), AAA servers, monitoring. Section 9 addresses these.

2. Design Methodology

The design follows the hierarchical model from Cisco's campus design guidance [1], which separates the network into core, distribution and access layers so that each layer has one job and failures stay local. For a 50-user, four-office client, a full three-tier build would add cost with no benefit. The core and distribution layers are therefore **collapsed** into one multilayer switch per city, the standard pattern for small and medium sites.

Table 3. *Hierarchical layer responsibilities*

Layer	Device (per site)	Responsibility
WAN edge	Cisco 2911 ISR	Inter-city routing over the serial WAN; RIPv2 neighbour to the remote site
Core / Distribution (collapsed)	Cisco 3560-24PS multilayer switch	Inter-VLAN routing (SVIs), DHCP server, RIPv2, trunk termination, SSH management
Access	2 × Cisco 2960-24TT	Endpoint ports (access VLAN, PortFast), 802.1Q uplink, wireless AP attachment
Endpoints	16 PCs + 4 laptops + 2 APs	Office users; laptops associate to a per-office WPA2-PSK SSID

2.1 Key design decisions

Table 4. *Design decisions and trade-offs*

Decision	Alternative considered	Rationale
Route between VLANs on the L3 core	Router-on-a-stick	Hardware (ASIC) switching at wire speed; keeps the WAN router free for inter-site traffic and removes a single-trunk bottleneck
RIPv2 (classless)	Static routes; OSPF	Meets the brief for dynamic routing with minimal configuration; supports VLSM. OSPF is recommended for production (§9)
VLSM /26 in London	One /24 per VLAN	Shows efficient address use: two 62-host LANs plus a /30 fit inside one /24 with 120 addresses kept in reserve
/24 per VLAN in Manchester	/26 as in London	Uses a /16 so each office can grow to 254 hosts without renumbering
/30 on point-to-point links	/24 or /31	Exactly two usable hosts; /31 (RFC 3021) is valid in production but less widely supported in labs
SSHv2 with local users	Telnet; console-only	Encrypted management traffic (RFC 4253); Telnet is plaintext and is disabled on every VTY line

3. Network Architecture

Figure 1 shows the logical architecture as built. It was redrawn from the device configurations rather than copied from the simulator canvas, so every interface, subnet and VLAN shown matches what is running on the devices.

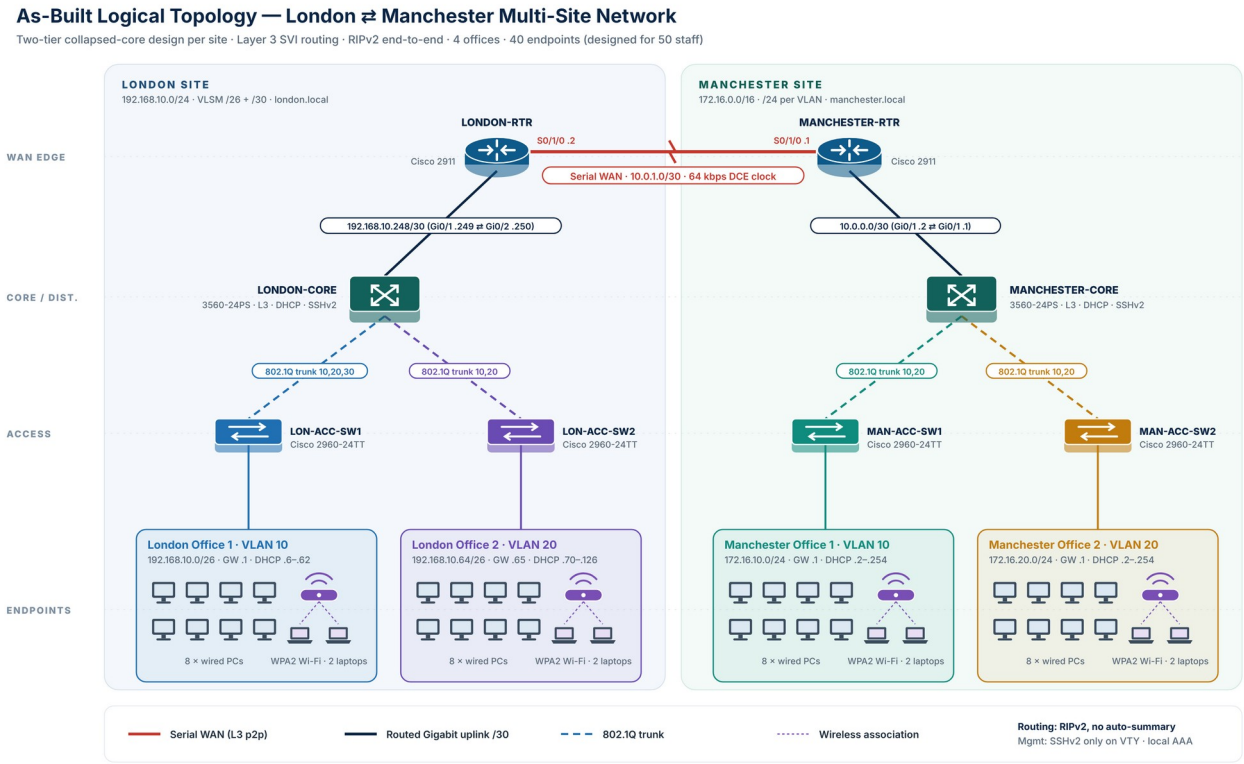


Figure 1. *As-built logical topology. Every address, interface and trunk shown is taken from the device running-configurations.*

3.1 Packet Tracer implementation

Figures 2 to 4 show the physical build in Cisco Packet Tracer. Each office is grouped into labelled “cabins” of PCs, with a wireless access point serving two laptops.

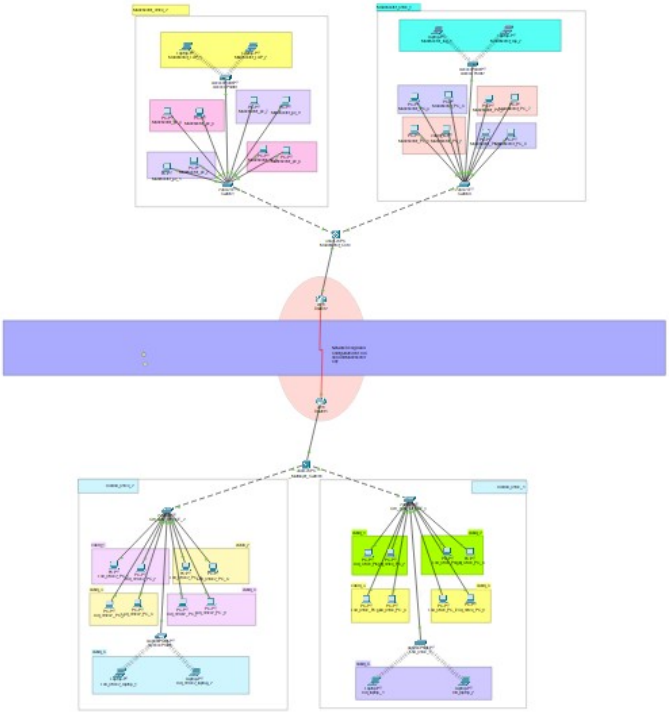


Figure 2. *Complete Packet Tracer workspace: London (top), the serial WAN (centre) and Manchester (bottom).*

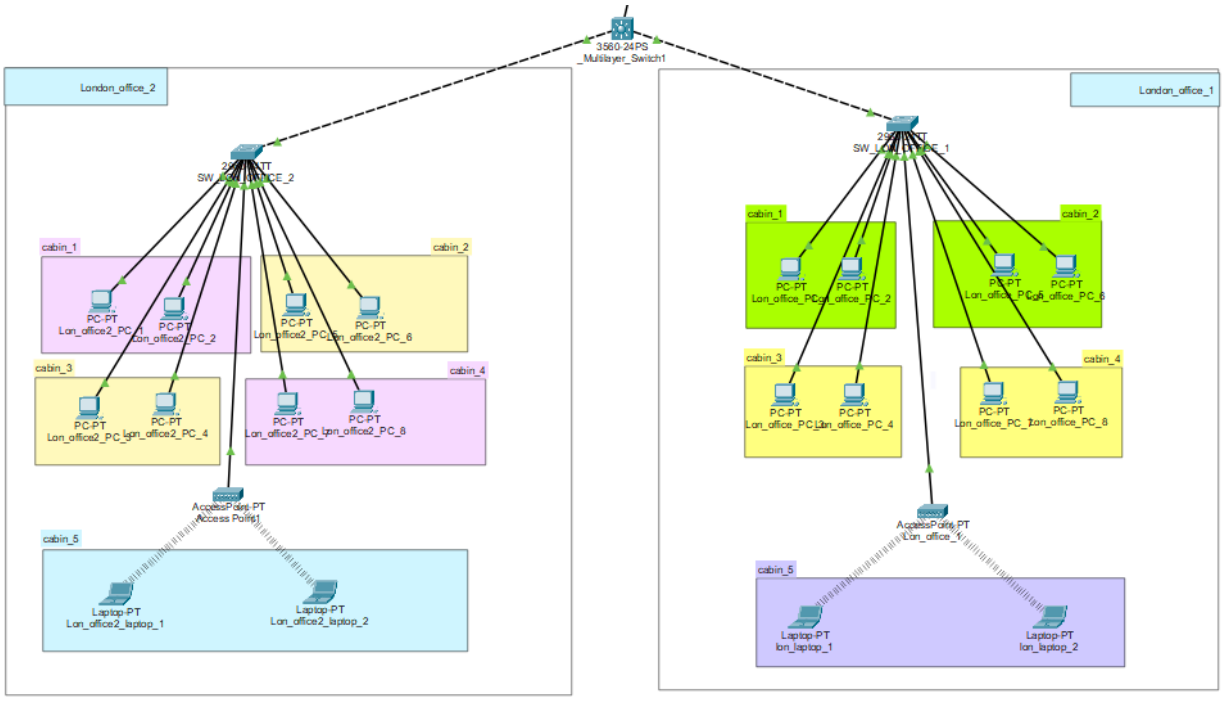


Figure 3. *London site: LONDON-CORE feeding Office 1 (VLAN 10) and Office 2 (VLAN 20) access switches.*

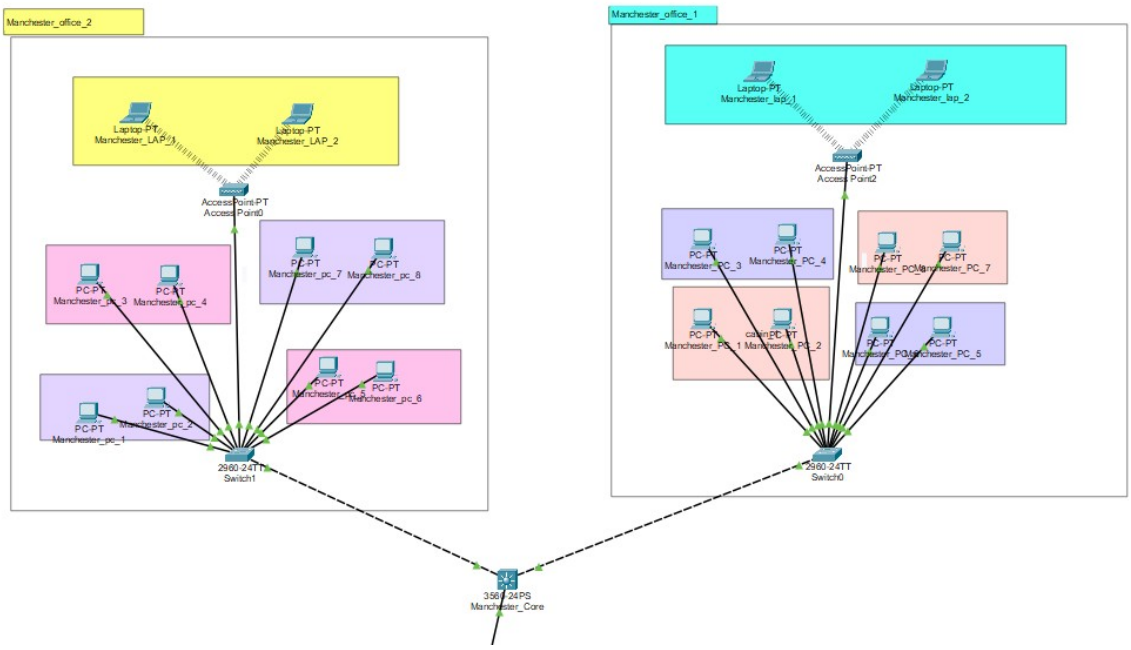


Figure 4. *Manchester site: MANCHESTER-CORE feeding Office 1 and Office 2 access switches, each with an access point.*

3.2 Device inventory

Table 5. *Device inventory (40 endpoints modelled for a 50-staff client)*

Role	Hostname	Model	Key interfaces
WAN router	London_Router	Cisco 2911	Gi0/1 192.168.10.249/30 · S0/1/0 10.0.1.2/30
WAN router (DCE)	Manchester_Router	Cisco 2911	Gi0/1 10.0.0.2/30 · S0/1/0 10.0.1.1/30, clock 64000
L3 core	LONDON-CORE	Cisco 3560-24PS	Gi0/2 .250/30 · SVI V10 .1 · SVI V20 .65
L3 core	Manchester_core	Cisco 3560-24PS	Gi0/1 10.0.0.1/30 · SVI V10/V20/V30 .1
Access	Lon_office_1 / Lon_office_2	Cisco 2960-24TT	Fa0/1 trunk · mgmt SVI .2 / .66
Access	manchester_office_1 / _2	Cisco 2960-24TT	Fa0/1 trunk · mgmt SVI 172.16.10.2 / 172.16.20.2
Wireless	4 × AccessPoint-PT	Generic AP	One SSID per office, WPA2-PSK
Endpoints	32 PCs, 8 laptops	PC-PT / Laptop-PT	8 wired + 2 wireless per office, DHCP

> **Sizing note**
> The lab models **40 endpoints** (8 PCs and 2 laptops in each of the 4 offices). The addressing plan has room for the client's full 50 staff, with headroom in every subnet.

4. IP Addressing and VLAN Design

All addressing uses private RFC 1918 space [3] with classless (CIDR) subnetting [4]. The two cities deliberately use different approaches so the report can compare them: London uses **variable-length subnet masking** to pack three subnets into a single /24, while Manchester uses a **hierarchical /16** with a clean /24 per VLAN.

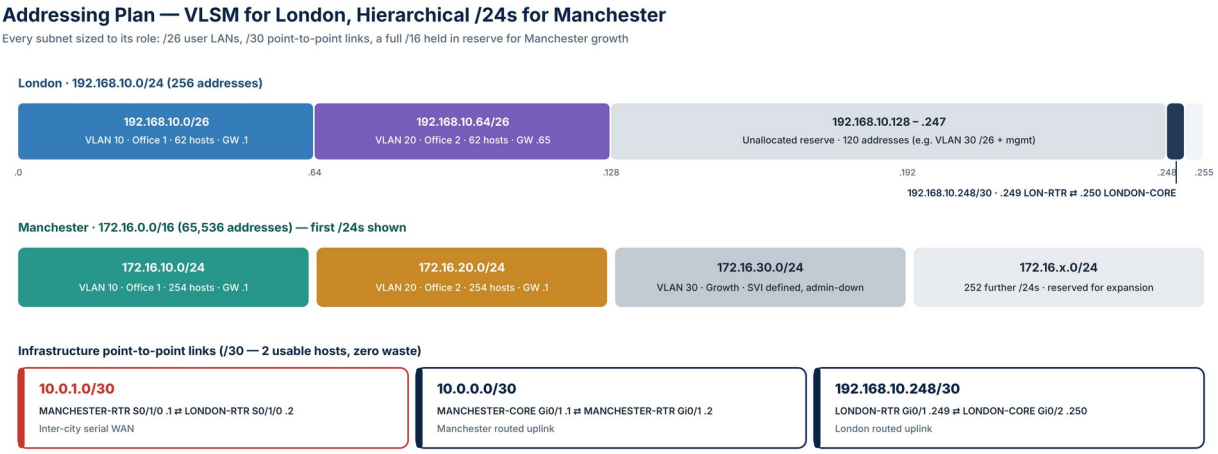


Figure 5. *Address-space utilisation. London uses VLSM inside 192.168.10.0/24; Manchester uses /24 blocks from 172.16.0.0/16.*

Table 6. *User subnets and DHCP scopes (taken from the running-configs)*

Subnet	VLAN / Site	Gateway (SVI)	Usable	Excluded	DHCP lease range
192.168.10.0/26	VLAN 10 · London Office 1	192.168.10.1	62	.1 – .5	.6 – .62
192.168.10.64/26	VLAN 20 · London Office 2	192.168.10.65	62	.65 – .69	.70 – .126
172.16.10.0/24	VLAN 10 · Manchester Office 1	172.16.10.1	254	.1	.2 – .254*
172.16.20.0/24	VLAN 20 · Manchester Office 2	172.16.20.1	254	.1	.2 – .254*
172.16.30.0/24	VLAN 30 · growth (SVI shut)	172.16.30.1	254	—	not yet enabled

* Each Manchester access switch has a management SVI at .2 that is **not** excluded from its pool. Section 7 records this as a DHCP conflict risk and gives the fix.

Table 7. *Point-to-point infrastructure links*

Link	Subnet	Side A	Side B
Inter-city serial WAN	10.0.1.0/30	Manchester_Router S0/1/0 · .1 (DCE)	London_Router S0/1/0 · .2
Manchester uplink	10.0.0.0/30	Manchester_core Gi0/1 · .1	Manchester_Router Gi0/1 · .2
London uplink	192.168.10.248/30	London_Router Gi0/1 · .249	LONDON-CORE Gi0/2 · .250

4.1 Subnetting rationale

- **/26 for London offices:** $2^6 - 2 = 62$ hosts. Each office needs 10 endpoints, a switch management address and the gateway, which leaves about 50 addresses of headroom per office and makes good use of the space.
- **/30 for links:** $2^2 - 2 = 2$ hosts, exactly one per router interface. No addresses are wasted on transit links.
- **Manchester /16:** holds 256 possible /24s. New offices or VLANs (voice, guest, management) can be added later without renumbering.

4.2 VLAN design

VLAN IDs 10 and 20 are used in **both** cities. This is safe because VLANs only exist inside one Layer 2 domain. The cities are joined by a routed link, so London's VLAN 10 and Manchester's VLAN 10 are separate broadcast domains that share a number. Using the same IDs for the same role at each site is common practice and makes configurations repeatable. Trunks are pruned with `switchport trunk allowed vlan`, so each link carries only the VLANs it needs (IEEE 802.1Q [5]).

5. Implementation

This section shows the configuration that delivers each function, using short excerpts taken from the devices. Full `show running-config` screenshots for all eight devices are in Appendix A, and the complete text configurations are in the project repository.

5.1 Inter-VLAN routing on the Layer 3 core

```
LONDON-CORE
ip routing
!
interface Vlan10
 ip address 192.168.10.1 255.255.255.192
interface Vlan20
 ip address 192.168.10.65 255.255.255.192
!
interface GigabitEthernet0/2
 no switchport                      ! routed port toward London_Router
 ip address 192.168.10.250 255.255.255.252
```

5.2 Centralised DHCP

```
LONDON-CORE
ip dhcp excluded-address 192.168.10.1 192.168.10.5
ip dhcp excluded-address 192.168.10.65 192.168.10.69
!
ip dhcp pool SW1-DHCP
 network 192.168.10.0 255.255.255.192
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20_POOL
 network 192.168.10.64 255.255.255.192
 default-router 192.168.10.65
 dns-server 8.8.8.8
```

5.3 Trunking and access ports

```
LONDON-CORE
interface FastEthernet0/1
 description Uplink to SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
```

```
Lon_office_1 (access switch)
interface FastEthernet0/2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast             ! edge port: skip listening/learning
```

5.4 RIPv2 dynamic routing

RIPv2 [2] runs on all four Layer 3 nodes. `no auto-summary` is essential because RIP otherwise summarises at classful boundaries. Without it, the /26 and /30 subnets of 192.168.10.0/24 would be advertised as a single /24, and the 172.16.x.0/24 networks as 172.16.0.0/16, which hides the real topology.

```
Manchester_core
router rip
 version 2
 network 10.0.0.0
 network 172.16.0.0
 no auto-summary
```

```
Manchester_Router
interface Serial0/1/0
 description TO_LONDON_ROUTER
 ip address 10.0.1.1 255.255.255.252
 clock rate 64000                    ! DCE end supplies the clock
```

5.5 Secure management (SSHv2)

```
Manchester_core
hostname Manchester_core
ip domain-name manchester.local
username admin secret <redacted>
crypto key generate rsa              ! one-time; keys are not shown in running-config
ip ssh version 2
banner motd ^ Unauthorized Access Prohibited ^
!
line vty 0 4
 exec-timeout 5 0
 login local
 transport input ssh                 ! Telnet disabled
```

6. Verification and Testing

Testing moved from the bottom of the stack to the top: Layer 3 reachability first, then the end-to-end path, then the services that depend on it (SSH, routing protocol state, DHCP). Each test has a pass criterion that was set before it was run.

Table 8. *Verification test matrix*

#	Test	Source → Destination	Pass criterion	Result
T1	Core-to-core reachability	LONDON-CORE ⇄ Manchester_core	ICMP success both directions	PASS · 5/5 (Figs. 6, 10)
T2	Inter-city host ping	192.168.10.73 ⇄ 172.16.10.7	0 % loss after ARP resolves	PASS · 4/4, TTL 124 (Fig. 7)
T3	Path trace (simulation)	Lon_office2_PC_5 → Manchester_PC_5	Expected 7-hop L2/L3 path	PASS (Fig. 8)
T4	SSH London → Manchester	Lon_office_PC_7 → 172.16.10.1	Login with local user, banner shown	PASS (Fig. 10)
T5	SSH Manchester → London	Manchester_PC_6 → 192.168.10.65	Login and privileged exec	PASS (Fig. 11)
T6	Routing convergence	`show ip route` on both cores	Every remote prefix present as R	PASS (Figs. 12–13)
T7	DHCP leasing	PCs in both cities	Correct IP, mask and gateway	PASS (Figs. 14–15)

6.1 Reachability and end-to-end path

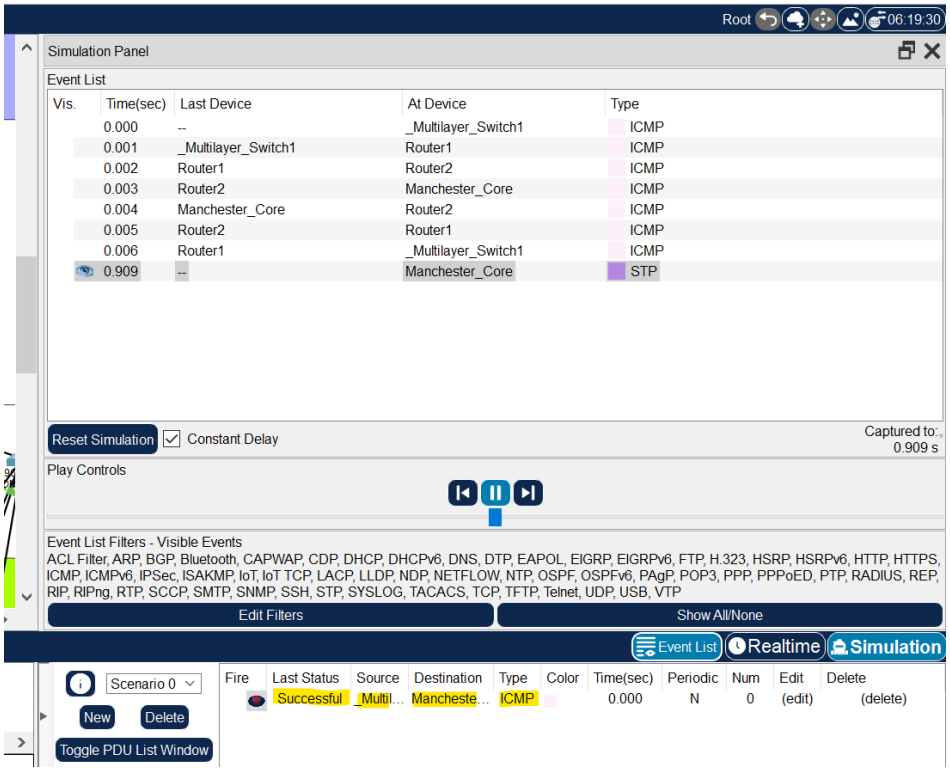


Figure 6. *T1: simulation of an ICMP echo from LONDON-CORE to Manchester_core, crossing both routers and returning successfully.*

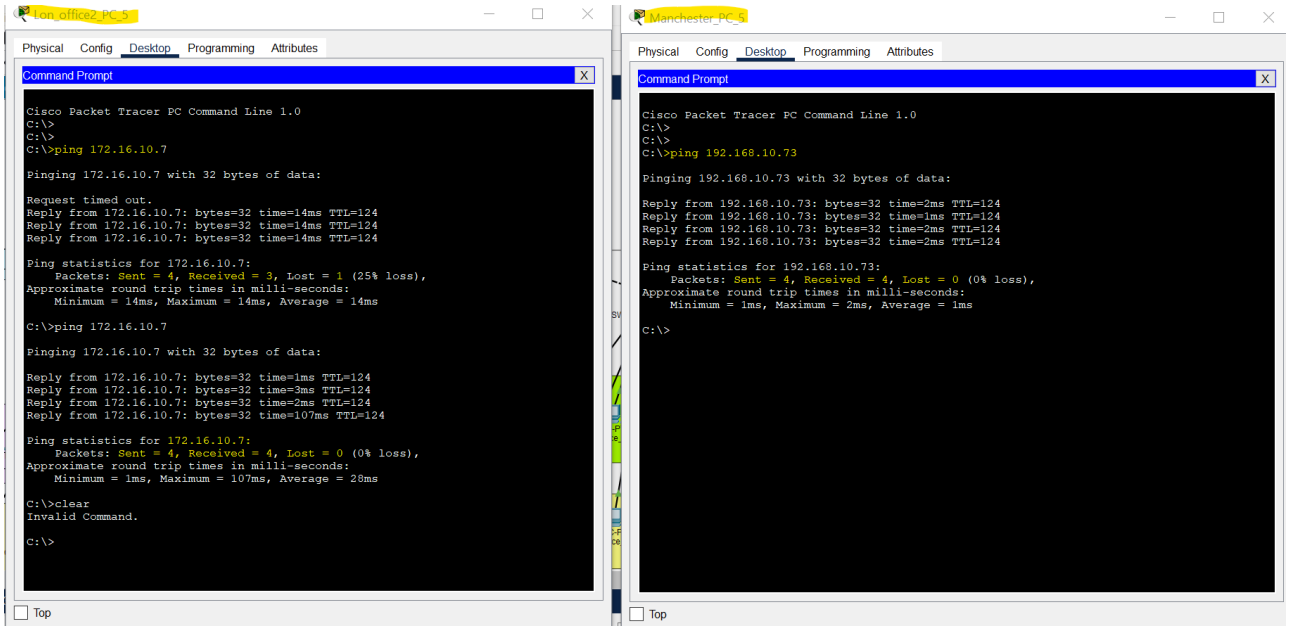


Figure 7. *T2: London Office 2 PC (192.168.10.73) ⇄ Manchester Office 1 PC (172.16.10.7). The single initial timeout is ARP resolution; all later replies succeed.*

The return TTL of **124** confirms the path. Windows-style hosts send with TTL 128, and four Layer 3 devices each decrement it by one (both cores and both routers), which gives 124 on arrival. Figure 8 shows the same path hop by hop, and Figure 9 explains the arithmetic.

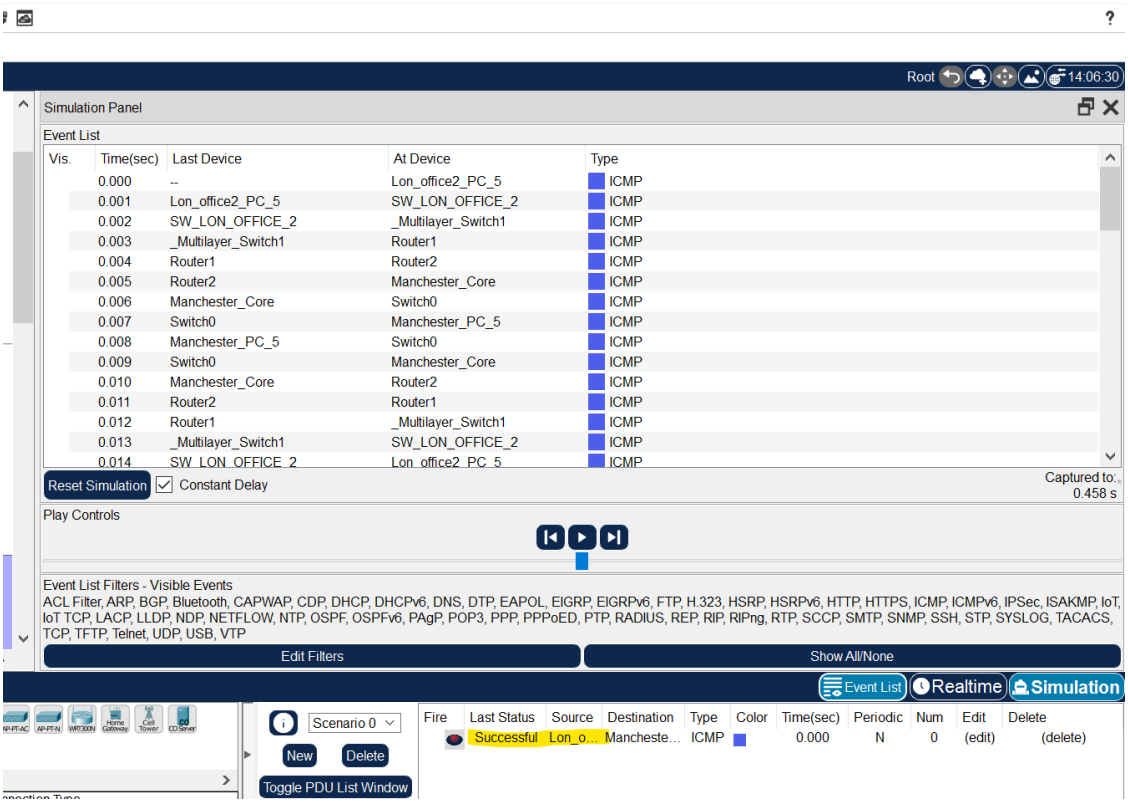


Figure 8. *T3: Packet Tracer event list for the ICMP flow PC → SW_LON_OFFICE_2 → LONDON-CORE → London_Router → Manchester_Router → Manchester_Core → Switch0 → Manchester_PC_5, then back.*

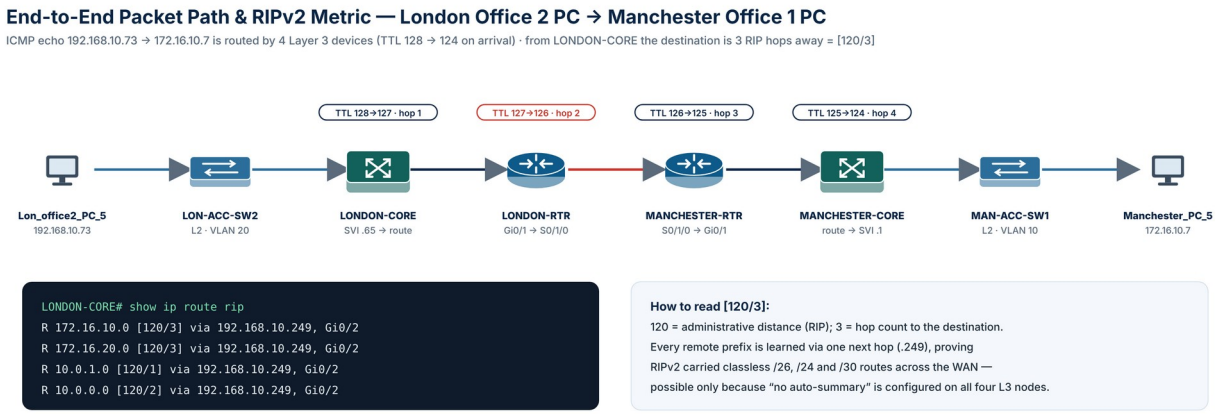


Figure 9. *Annotated packet path and RIP metric interpretation.*

6.2 Secure remote management

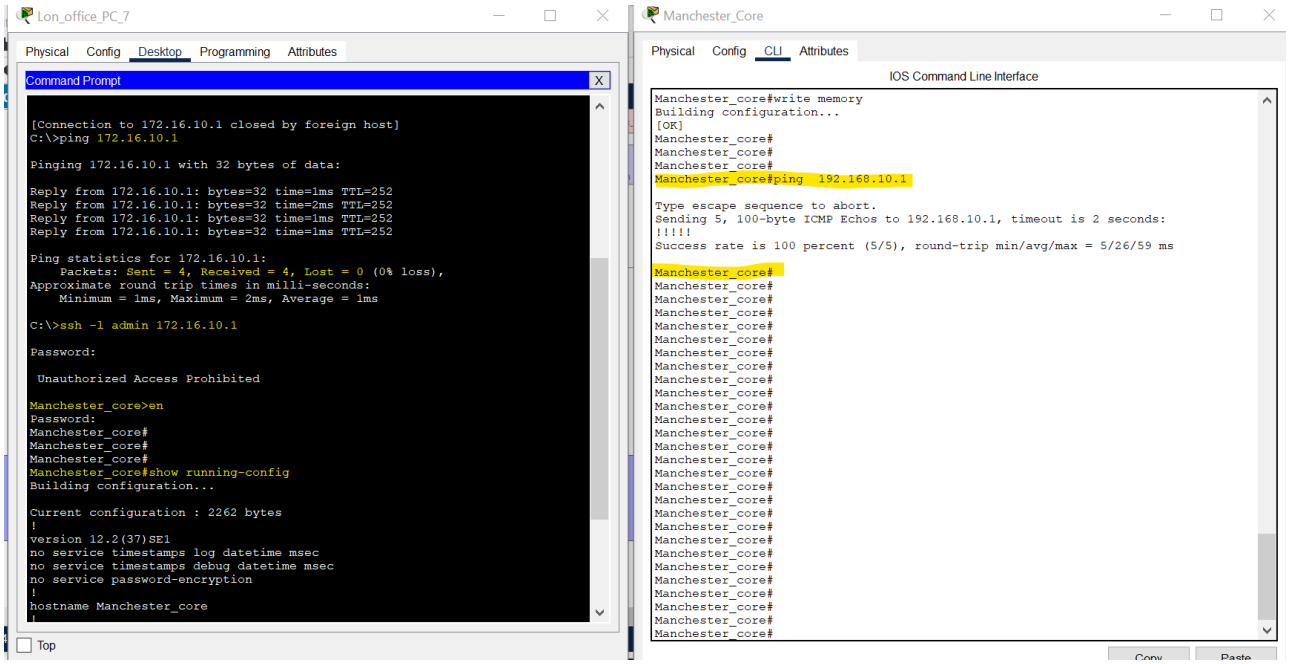


Figure 10. *T4: SSH from London Office 1 PC to Manchester_core (172.16.10.1). Local authentication, MOTD banner and privileged exec all work.*

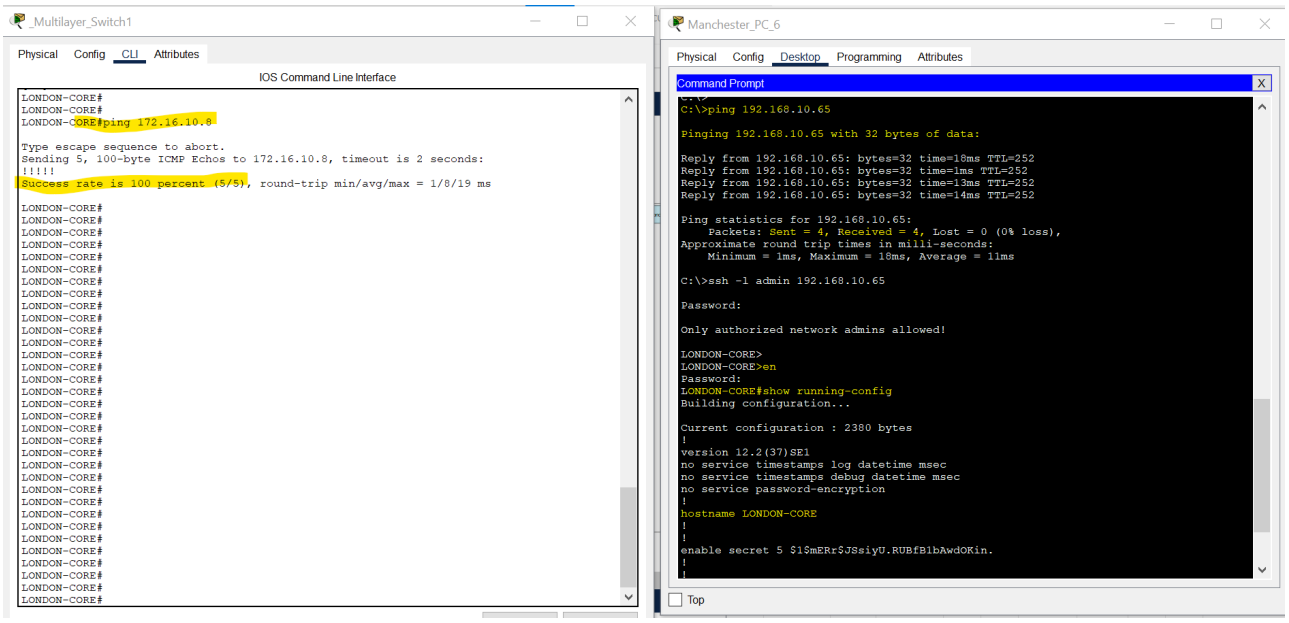


Figure 11. *T5: SSH from Manchester_PC_6 to LONDON-CORE (192.168.10.65), alongside a successful core-to-host ping (5/5).*

6.3 Routing-table convergence

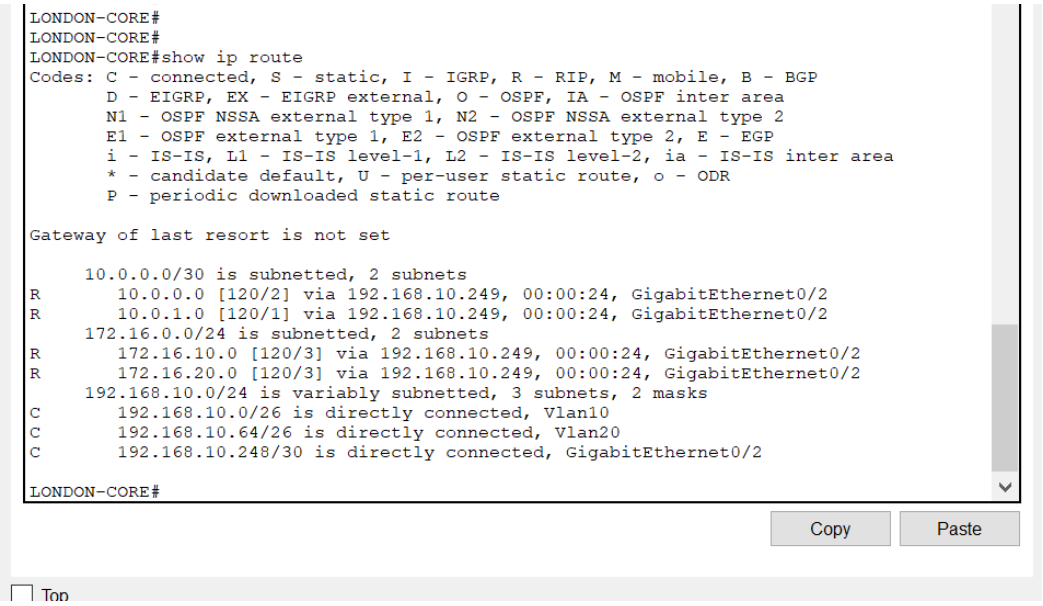


Figure 12. *T6: LONDON-CORE routing table. Manchester VLANs are learned at metric 3, the WAN /30 at metric 1 and the Manchester uplink at metric 2.*

```
Manchester_core>
Manchester_core>en
Password:
Password:
Manchester_core#
Manchester_core#
Manchester_core#
Manchester_core# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, GigabitEthernet0/1
R       10.0.1.0 [120/1] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.10.0 is directly connected, Vlan10
C       172.16.20.0 is directly connected, Vlan20
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
R       192.168.10.0/26 [120/3] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
R       192.168.10.64/26 [120/3] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
R       192.168.10.248/30 [120/2] via 10.0.0.2, 00:00:26, GigabitEthernet0/1

Manchester_core#
Manchester_core#
```

Copy Paste

Top

Figure 13. *T6: Manchester_core routing table. Both London /26 LANs and the London /30 are learned via 10.0.0.2.*

Both tables show `192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks`, which proves that the classless /26 and /30 prefixes crossed the WAN without being summarised. The metrics match the physical hop counts in Figure 1, which is a quick sanity check that RIP has learned the topology correctly.

6.4 DHCP

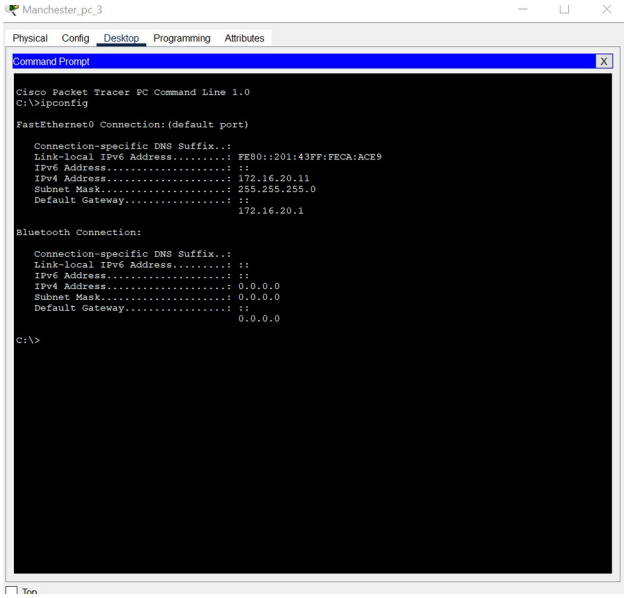


Figure 14. *T7: Manchester_pc_3 leased 172.16.20.11/24, gateway 172.16.20.1 (VLAN 20).*

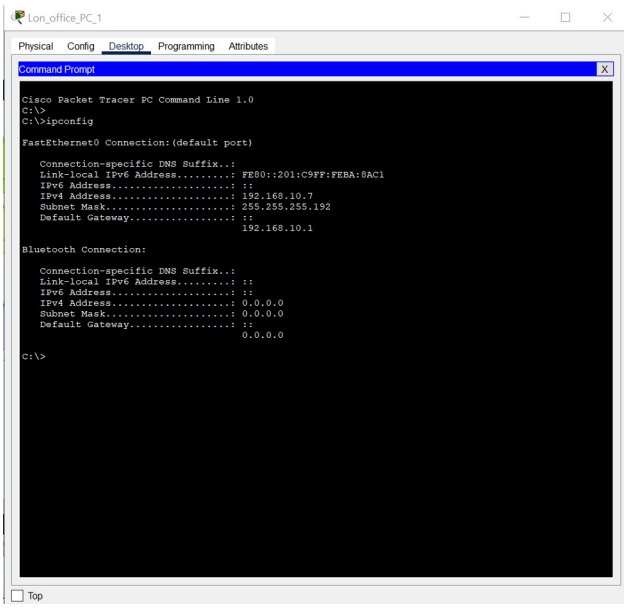


Figure 15. *T7: Lon_office_PC_1 leased 192.168.10.7/26, gateway 192.168.10.1 (VLAN 10).*

7. Security Assessment

The running configurations were reviewed against the CIS Cisco IOS Benchmark [8] and NIST wireless guidance [9]. The lab already has a good baseline. The gaps below are listed openly because this is the review a real deployment would go through before go-live.

7.1 Controls implemented

- SSH-only VTY access (`transport input ssh`) with local usernames on all eight network devices, so Telnet is disabled. `ip ssh version 2` is enforced on six of the eight.
- `enable secret` (hashed) instead of `enable password`; legal MOTD banners on core and access switches.
- 5-minute VTY idle timeout on Manchester_core; `no ip domain-lookup` to stop delays from mistyped commands.
- Default VLAN 1 SVI shut down; trunks pruned to required VLANs; PortFast on edge ports only.
- Unused access ports administratively shut on Lon_office_2 and both Manchester access switches.
- WPA2-PSK on every office SSID, with a separate SSID per office.

7.2 Findings and remediation

Table 9. *Configuration audit findings*

#	Finding (observed in config)	Risk	Recommended fix
F1	Console password stored in plaintext (`password london`) and `no service password-encryption` on LONDON-CORE and London access switches	High	`service password-encryption`; console `login local`; remove line passwords
F2	Secrets use type 5 (MD5) hashing	Medium	Type 9 (scrypt) or type 8 on platforms that support it
F3	Manchester DHCP pools do not exclude access-switch SVIs (.2)	Medium	`ip dhcp excluded-address 172.16.10.1 172.16.10.9` (and the same for .20.x)
F4	RIP updates are unauthenticated; a rogue router could inject routes	High	RIPv2 MD5 key-chain authentication, or move to OSPF with authentication
F5	No ACLs: any VLAN can reach every device's management plane	High	VTY `access-class` allowing only a management subnet; inter-VLAN ACLs
F6	No Layer 2 protections (port-security, DHCP snooping, DAI, BPDU guard)	Medium	Enable on all access ports; set an explicit STP root on each core
F7	Single 64 kbps serial WAN is a single point of failure and a bottleneck	High	Dual WAN (MPLS + IPsec VPN) with fast failover (see §9)
F8	Manchester core trunks rely on DTP negotiation; native VLAN 1; `ip ssh version 2` missing on LONDON-CORE and Lon_office_1	Low	`switchport mode trunk`, `switchport nonegotiate`, unused native VLAN, enforce SSHv2 everywhere
F9	DNS points to 8.8.8.8 but the network has no internet edge	Low	Internal DNS, or add an edge router/firewall with NAT

8. Challenges and Resolutions

Table 10. *Issues encountered during the build*

Challenge	Root cause	Resolution
RIP routes slow to appear	Default RIP timers: 30 s updates and 180 s invalid timer	Verified with `debug ip rip` and allowed time to converge; timers tuned during testing
SSH authentication failures	RSA keys generated before the domain name was set and consistent	Set `ip domain-name`, regenerated keys, confirmed `ip ssh version 2`
DHCP address conflicts	Pool overlapped statically assigned addresses	Added exclusion ranges; checked with `show ip dhcp conflict`
Hosts in the wrong VLAN	Access ports left in VLAN 1	Corrected with `switchport access vlan 20`; verified with `show vlan brief`
First ping times out	ARP resolution along a multi-hop path	Expected behaviour; later pings succeed (Fig. 7)

9. From Lab to Production

The lab meets the brief, but a production network for the same client would be held to stricter standards for availability, security and visibility. Figure 16 shows the recommended target state. It keeps the same addressing and VLAN logic so that migration is incremental, and adds redundancy and security at every tier.

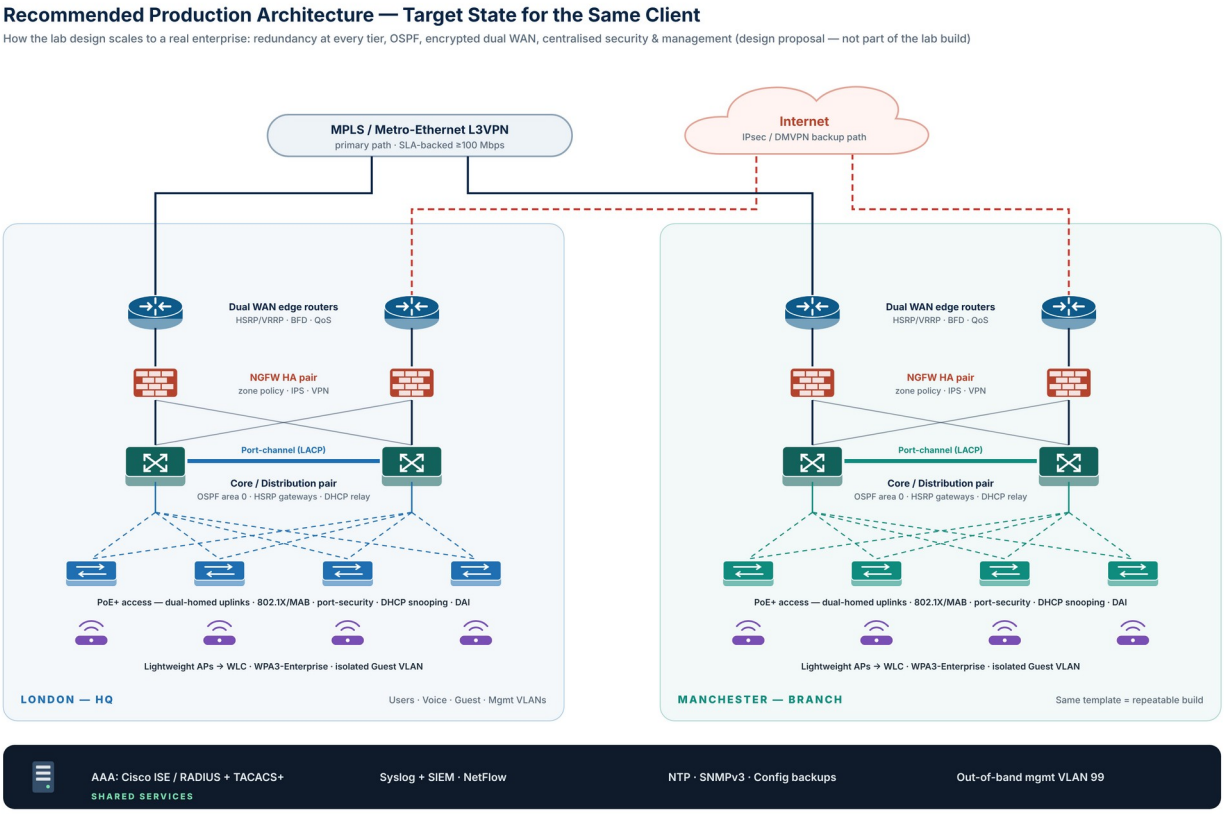


Figure 16. *Recommended production architecture (design proposal; not part of the lab build).*

Table 11. *Phased production roadmap*

Phase	Change	Benefit
1 · Harden (days)	Fix F1–F6: encrypted secrets, VTY ACLs, RIP/OSPF authentication, port-security, DHCP snooping, DAI, BPDU guard	Closes the high-risk audit findings at near-zero cost
2 · Resilience (weeks)	OSPF area 0 [7] replacing RIP; second core per site with HSRP; LACP uplinks; dual WAN (MPLS primary, IPsec/DMVPN backup)	Sub-second failover, no single point of failure, scales past RIP's 15-hop limit
3 · Security edge	NGFW HA pair per site with zone-based policy [10]; NAT and internet breakout; guest VLAN isolated	Controlled north–south traffic and safe internet access
4 · Identity & wireless	802.1X / MAB via RADIUS (Cisco ISE); TACACS+ for admins; WLC-managed APs with WPA3-Enterprise	Per-user access control and an accountable admin audit trail
5 · Visibility	Syslog to SIEM, NetFlow, SNMPv3, NTP, automated config backups, management VLAN 99	Faster troubleshooting, incident response and compliance evidence

10. Conclusion

The project delivered a working, verified multi-site network that meets all seven client requirements. A hierarchical design keeps each office's traffic local, VLSM and /30 links use address space efficiently, Layer 3 cores route between VLANs at hardware speed, RIPv2 carries classless routes across the WAN, and SSHv2 protects every management session. All seven verification tests passed.

The report also looks critically at its own design. The configuration audit identified nine improvements, and the phased roadmap shows how the same design would become a resilient, secure production network. That is the gap between building something that works and building something an organisation can rely on.

References

[1] Cisco Systems, "Campus LAN and Wireless LAN Solution Design Guide," Cisco Validated Design.
[2] G. Malkin, "RIP Version 2," IETF RFC 2453, 1998.
[3] Y. Rekhter et al., "Address Allocation for Private Internets," IETF RFC 1918, 1996.
[4] V. Fuller and T. Li, "Classless Inter-domain Routing (CIDR)," IETF RFC 4632, 2006.
[5] IEEE Std 802.1Q, "Bridges and Bridged Networks," IEEE.
[6] T. Ylonen and C. Lonvick, "The Secure Shell (SSH) Transport Layer Protocol," IETF RFC 4253, 2006.
[7] J. Moy, "OSPF Version 2," IETF RFC 2328, 1998.
[8] Center for Internet Security, "CIS Cisco IOS Benchmark."
[9] NIST SP 800-153, "Guidelines for Securing Wireless Local Area Networks (WLANs)," 2012.
[10] NIST SP 800-41 Rev. 1, "Guidelines on Firewalls and Firewall Policy," 2009.
[11] R. Droms, "Dynamic Host Configuration Protocol," IETF RFC 2131, 1997.

Appendix A — Configuration Evidence

Running-configuration captures for all eight network devices, grouped by site. Plain-text versions of the same configurations (with secrets redacted) are in the project repository.

London — LONDON-CORE (3560)

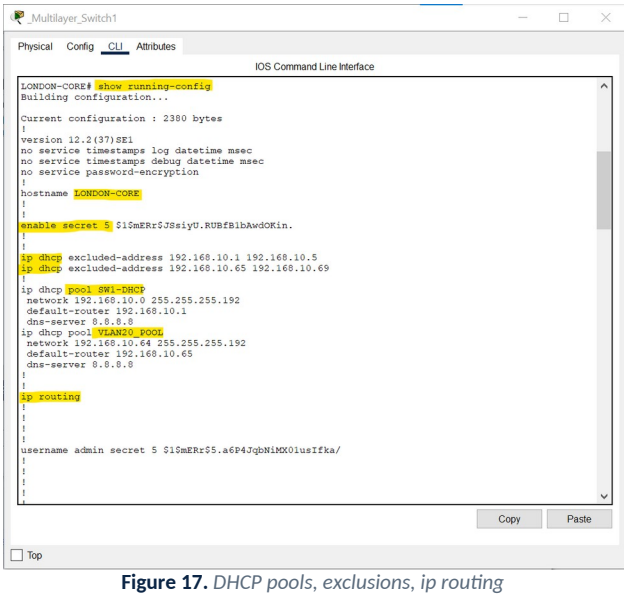


Figure 17. *DHCP pools, exclusions, ip routing*

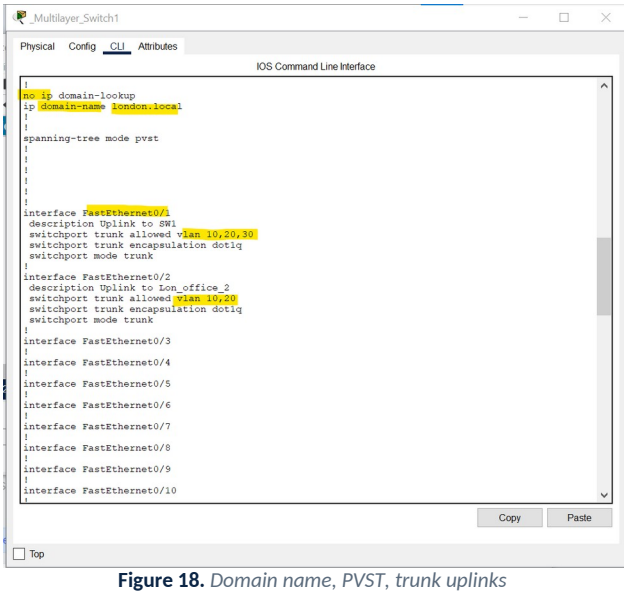


Figure 18. *Domain name, PVST, trunk uplinks*

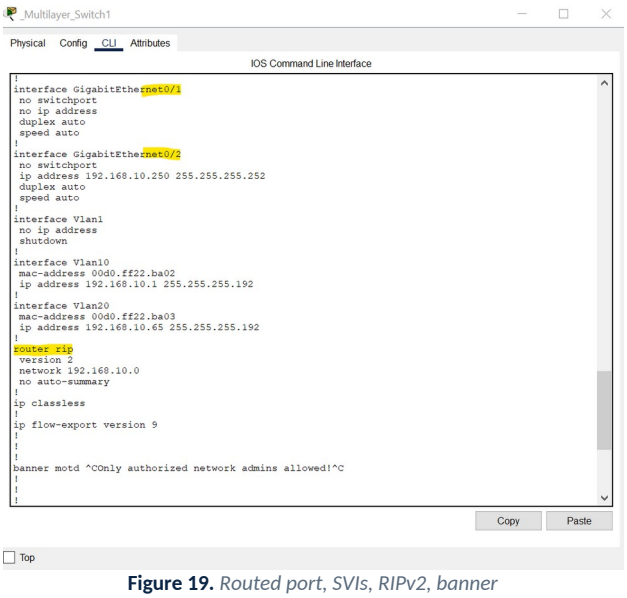


Figure 19. *Routed port, SVIs, RIPv2, banner*

London — access switches (2960)

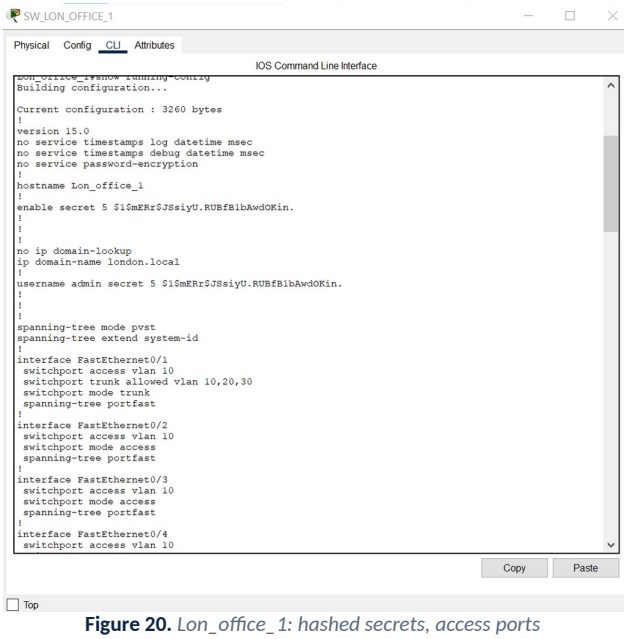


Figure 20. *Lon_office_1: hashed secrets, access ports*

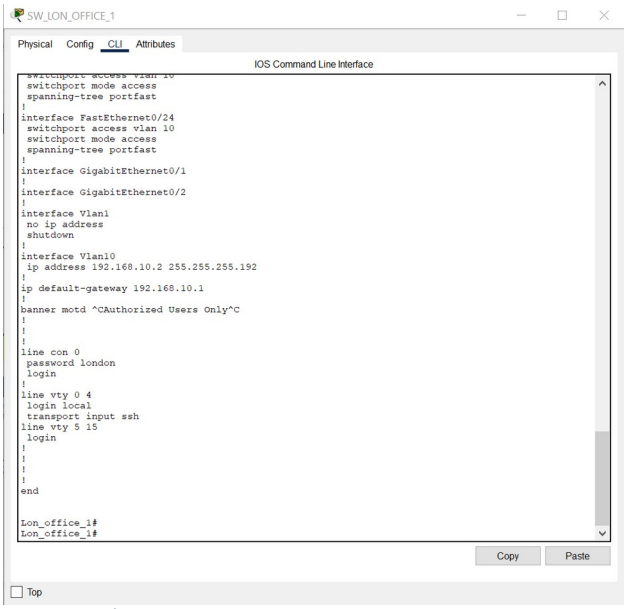


Figure 21. *Lon_office_1: mgmt SVI, SSH-only VTY*

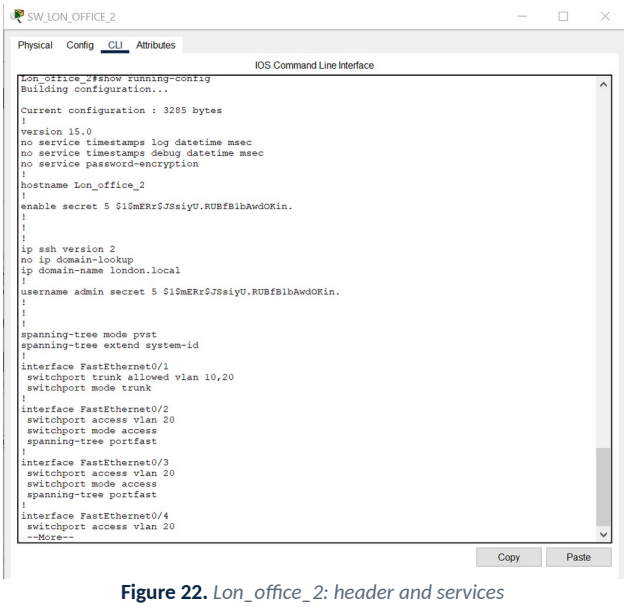


Figure 22. *Lon_office_2: header and services*

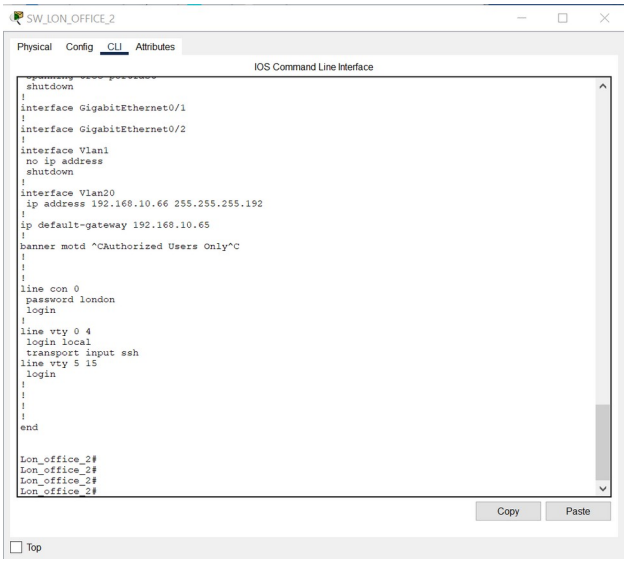


Figure 23. *Lon_office_2: mgmt SVI VLAN 20*

London — London_Router (2911)

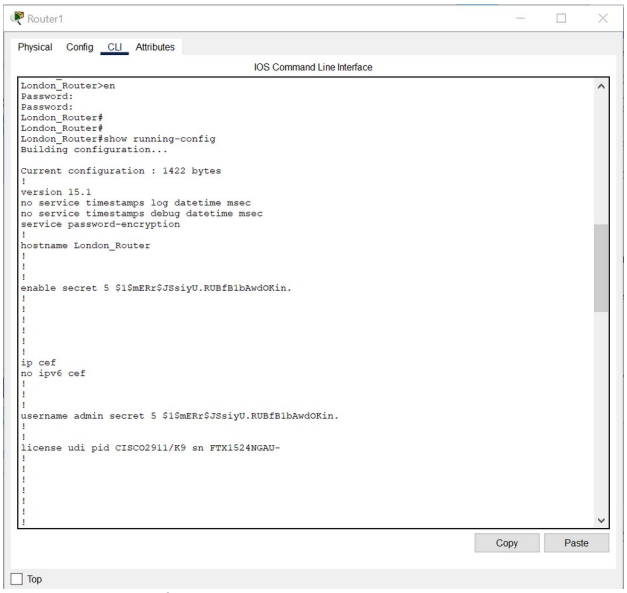


Figure 24. *Secrets, local user, license*

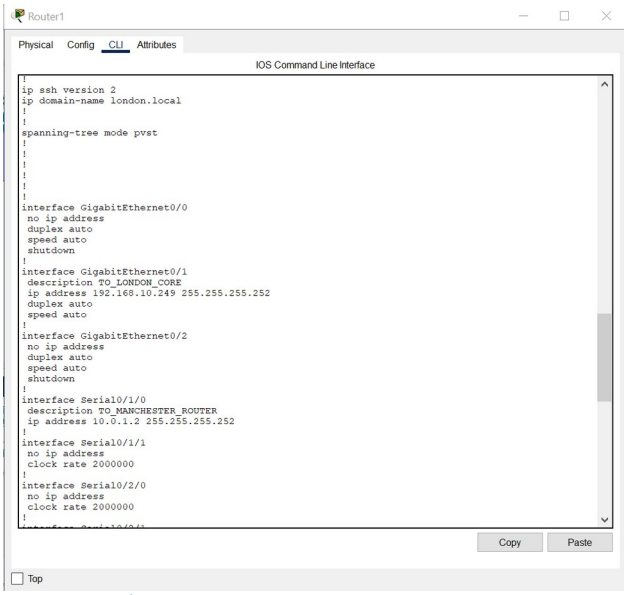


Figure 25. *SSHv2, Gi0/1 to core, S0/1/0 WAN*

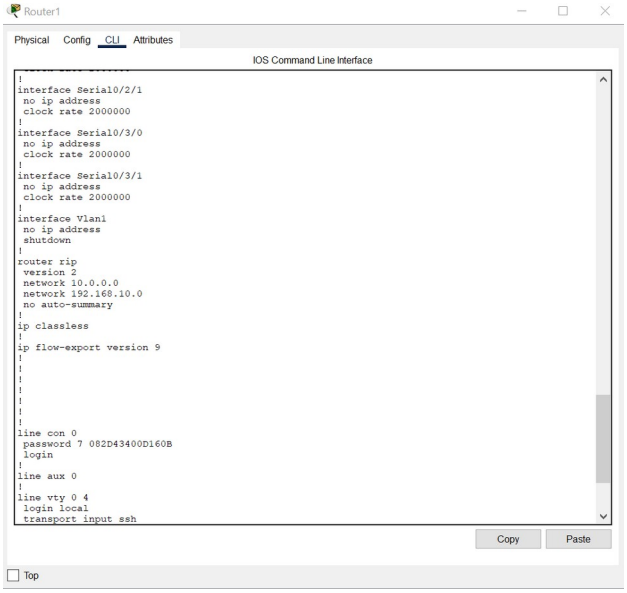


Figure 26. *RIPv2 networks, VTY SSH-only*

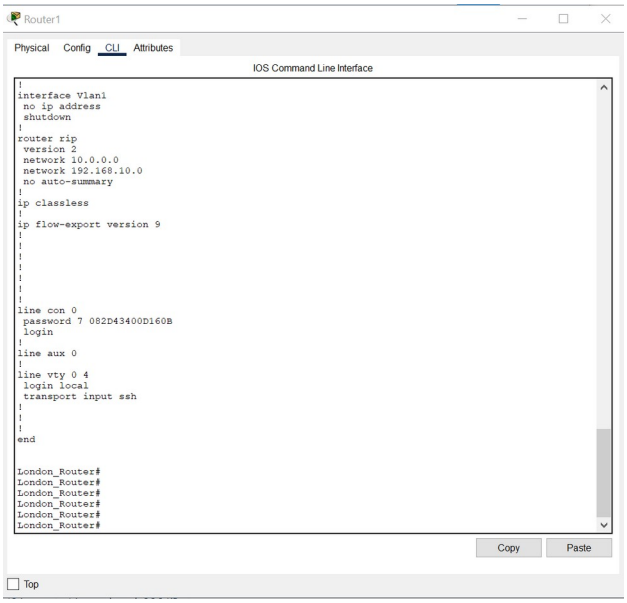


Figure 27. *End of configuration*

Manchester — Manchester_core (3560)

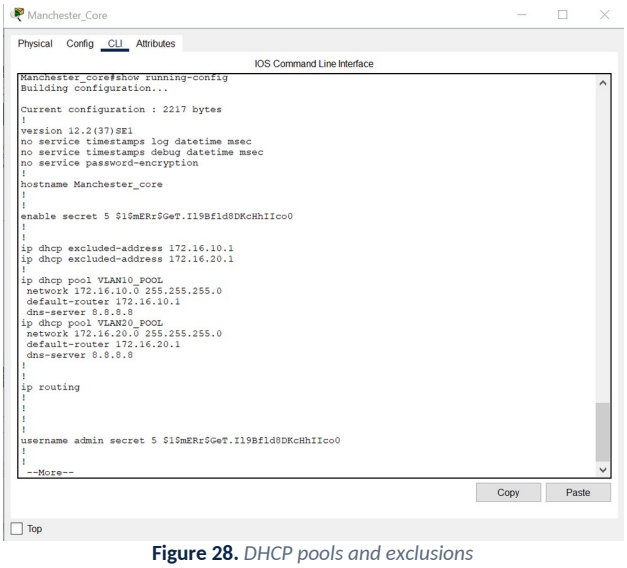


Figure 28. *DHCP pools and exclusions*

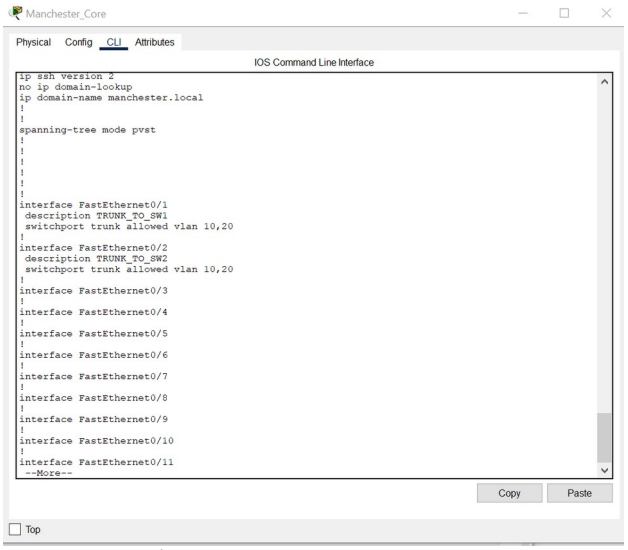


Figure 29. *SSHv2, domain, trunk ports*

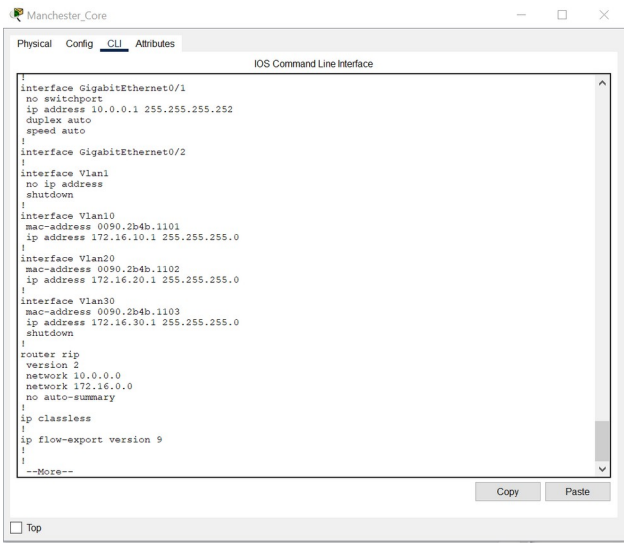


Figure 30. *Routed Gi0/1, SVIs incl. VLAN 30, RIPv2*

Manchester — Manchester_Router (2911)

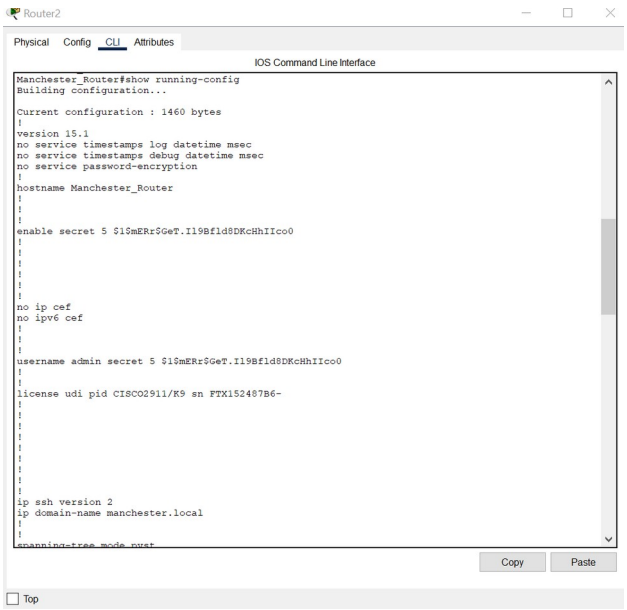


Figure 31. *Secrets, local user, SSHv2*

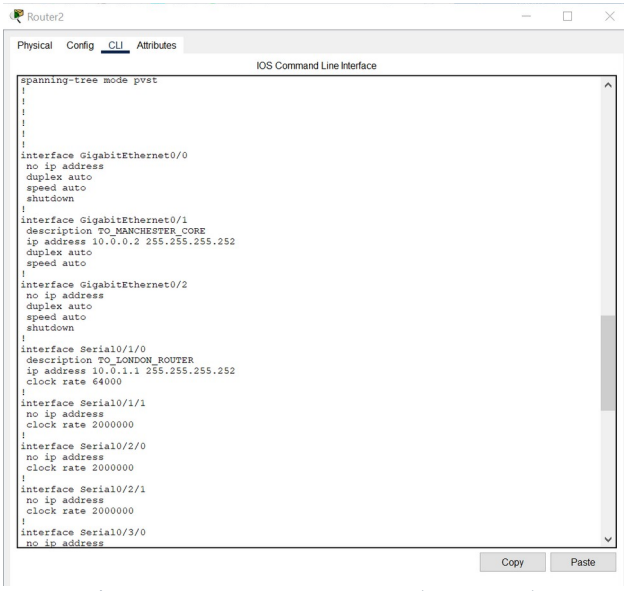


Figure 32. *Gi0/1 to core, S0/1/0 WAN (DCE 64 kbps)*

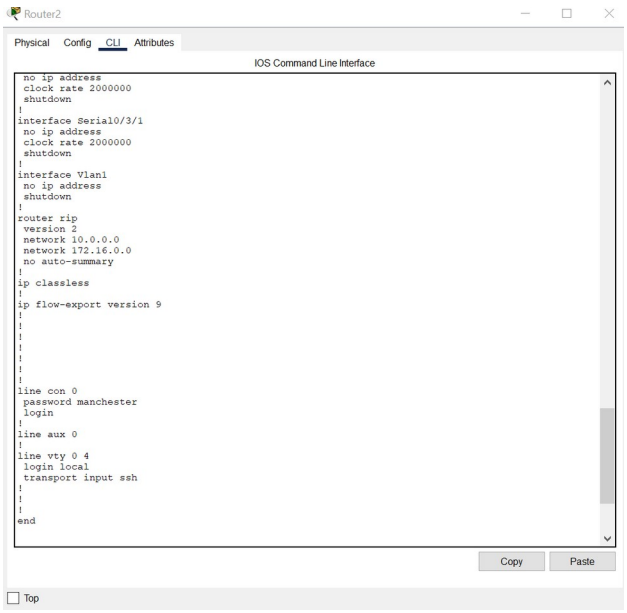


Figure 33. *Unused serials shut, VLAN 1 shut*

Manchester — access switches (2960)

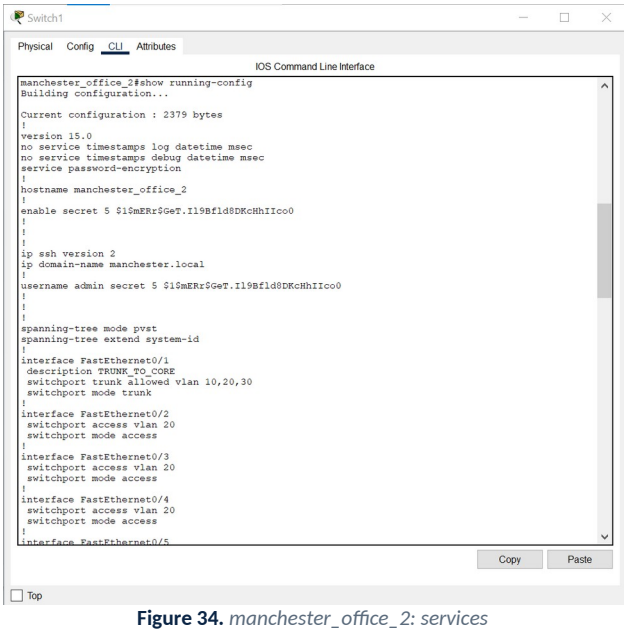


Figure 34. *manchester_office_2: services*

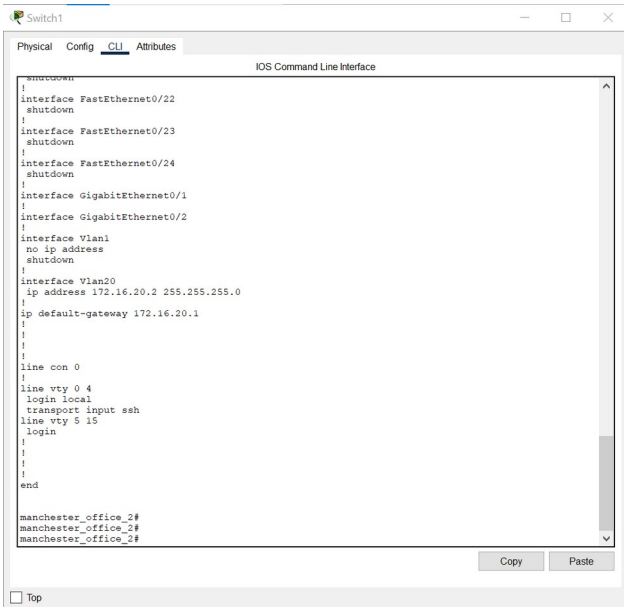


Figure 35. *manchester_office_2: ports and SVI*

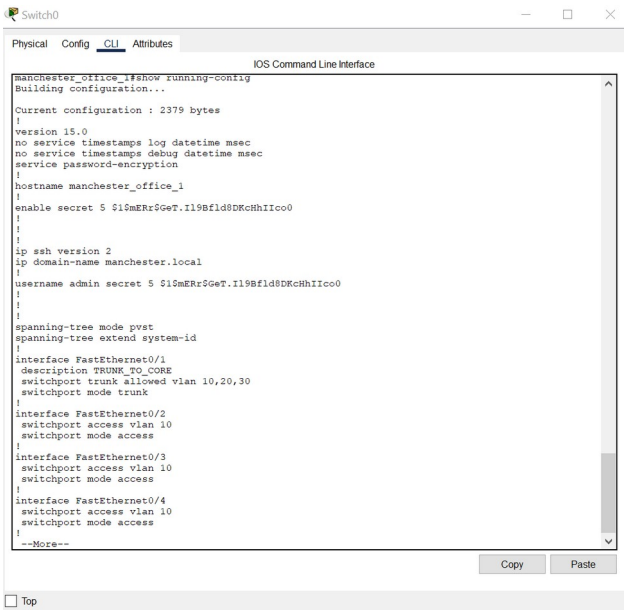


Figure 36. *manchester_office_1: services*

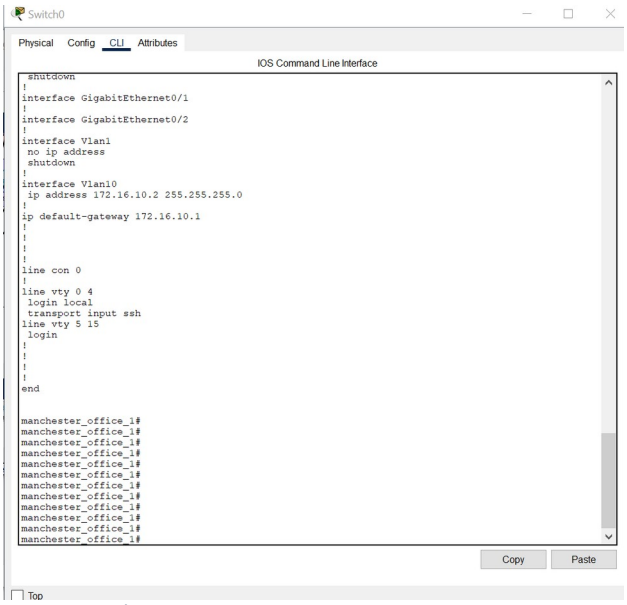


Figure 37. *manchester_office_1: ports and SVI*

Appendix B — Reproducing This Lab

1. Install Cisco Packet Tracer 8.2 or later (free through Cisco Networking Academy).
2. Open `London_Manchester_Network.pkt` from the repository.
3. Wait about 60 seconds for STP and RIP to converge (or press Fast Forward Time).
4. If a PC shows a 169.254.x.x address, open Desktop → IP Configuration and select DHCP again.
5. Run the tests in Table 8: `ping`, `ssh -l admin <SVI>`, and `show ip route` on each core.

> **Lab credentials**
> Credentials in the lab file are for simulation only and are deliberately simple. Never reuse lab credentials on real equipment.

About the Author

Muhammad Sohaib Rafiq is a final-year Cyber Security student at Air University, Islamabad. His work focuses on secure network design, malware analysis and network-based threat detection. This case study is part of a public portfolio that pairs design work with security auditing.

He welcomes technical feedback, collaboration and conversations about network engineering, SOC and security roles.

LinkedIn: [add your profile URL] · GitHub: [add your repository URL]